Self-starting Siphon Priming: Geometry, Dynamics, and Loss

Models

Self-starting siphon behaviour and priming have been discussed in the fluid-mechanics and physics-education literature; see, for example, Potter and Barnes [5], Jumper and Stanchev [6], Zheng *et al.* [7], Ortloff and Kassinos [8], and Sedlar *et al.* [9]. The slug-flow model developed below focuses on the transient filling and draining dynamics of a self-starting siphon, including frictional and minor losses. Throughout, we make the following modelling assumptions:

- The fluid is incompressible and Newtonian with constant density ρ and viscosity μ (e.g. water at room temperature).
- The pipe is rigid, stationary, and has a uniform circular cross-section of diameter D and area A; the centreline geometry is fixed.
- Flow in the pipe is treated as one-dimensional slug (plug) flow, so the mean velocity $v(t)$ is uniform over each cross-section.
- The flow remains single-phase liquid in the regions modelled; effects of cavitation, trapped air pockets, and surface tension are neglected.
- The container is large enough that the velocity at the free surface is negligible, so the container remains hydrostatic at each instant.
- Both the container free surface and the outlet discharge to the same atmospheric pressure, and the external air dynamics are neglected.

1 Part I: Geometry and Lossless Equation of Motion

1.1 Pipe Geometry

We describe the pipe centerline by a sequence of points

$$A(0, h_0), \quad B(x_1, h_1), \quad C(x_2, h_2), \quad D(x_3, h_3), \quad E(x_4, h_4),$$

where the horizontal coordinates are x_i and the vertical coordinates are h_i. The lengths of the straight segments between consecutive points are

$$\begin{aligned} L_1 &= |AB| = \sqrt{x_1^2 + (h_1 - h_0)^2}, \\ L_2 &= |BC| = \sqrt{(x_2 - x_1)^2 + (h_2 - h_1)^2}, \\ L_3 &= |CD| = \sqrt{(x_3 - x_2)^2 + (h_3 - h_2)^2}, \\ L_4 &= |DE| = \sqrt{(x_4 - x_3)^2 + (h_4 - h_3)^2}. \end{aligned}$$

The total pipe length is

$$L_{\max} = L_1 + L_2 + L_3 + L_4.$$

1.2 Height Along the Pipe

Let L denote the arc-length coordinate measured along the pipe from point A. We define a piecewise height function $h(L)$ along the pipe. For each straight segment we use a local parameter in $[0, 1]$; the functions h_1, h_2, h_3, h_4 map that local parameter to the physical height on each segment:

$$h(L) = \begin{cases} h_1\left(\dfrac{L}{L_1}\right), & 0 \le L \le L_1, \\ h_2\left(\dfrac{L - L_1}{L_2}\right), & L_1 \le L \le L_1 + L_2, \\ h_3\left(\dfrac{L - L_1 - L_2}{L_3}\right), & L_1 + L_2 \le L \le L_1 + L_2 + L_3, \\ h_4\left(\dfrac{L - L_1 - L_2 - L_3}{L_4}\right), & L_1 + L_2 + L_3 \le L \le L_{\max}. \end{cases}$$

If each segment is straight, then each h_i is an affine function of its local parameter.

1.3 Container Water Level

We choose a reference time t_0 at which

- the water plug in the pipe has length $L(t_0)$,
- the container water height (free surface) is $h_c(t_0)$,
- the pipe cross-sectional area is A,
- the container free-surface area is A_0.

Let $V_c(t)$ be the volume of water in the container *outside* the pipe walls. During the stage when water is still entering the pipe from the container, the volumetric flow rate into the pipe is

$$Q(t) = A\,\dot{L}_{\text{in}}(t),$$

where $\dot{L}_{\text{in}}$ is the rate at which the water *front* in the pipe moves away from the container. During the initial filling phase we will have $\dot{L}_{\text{in}} = \dot{L}(t)$; later, once the pipe is completely filled, we must treat L as constant and use a different description (see Part II).

For the container alone, mass conservation gives

$$\frac{\mathrm{d}V_c}{\mathrm{d}t} = -Q(t) = -A\,\dot{L}_{\text{in}}(t).$$

Integrating in time from the reference instant t_0 to a general time t and *during the period where* $\dot{L}_{in} = \dot{L}$, we get

$$V_c(t) = V_c(t_0) - A\,[L(t) - L(t_0)].$$

Since $V_c(t) = A_0 h_c(t)$, this implies

$$h_c(t) = h_c(t_0) - \frac{A}{A_0}\,[L(t) - L(t_0)]. \tag{1}$$

This form is completely general and does not assume any particular initial pattern of filled/empty segments. In particular, any portions of the pipe that are submerged in the container but initially empty (for example, all of L_2 and some part of L_3) simply modify the numerical values of $h_c(t_0)$ and $L(t_0)$, but do not change the linear relation

$$\frac{\mathrm{d}h_c}{\mathrm{d}L} = -\frac{A}{A_0}.$$

If we denote by V_0 the actual water volume in the container at $t = 0$, and by $L_0 = L(0)$ the actual water length in the pipe at $t = 0$, then $h_c(0) = V_0/A_0$ and eq. (1) can be written as

$$h_c(t) = \frac{V_0 - A\,[L(t) - L_0]}{A_0}. \tag{2}$$

Here V_0 must exclude any submerged but initially *empty* pipe interior; those volumes are simply not water at $t = 0$.

1.4 Lossless Equation of Motion: Filling Phase

We first derive the lossless equation of motion for the stage where the water plug is still advancing along the pipe, i.e.

$$L(t_0) < L(t) < L_{\max}.$$

We model the water inside the pipe as a slug of length L, cross-sectional area A, density ρ, and mass

$$m = \rho A L.$$

Let g be the gravitational acceleration, and neglect friction and other losses. Atmospheric pressure acts both at the free surface in the container and at the pipe outlet, so it cancels in the driving pressure difference.

During the filling phase we have the kinematic relation

$$v(t) = \dot{L}(t), \tag{3}$$

where $v(t)$ is the mean velocity in the pipe.

We will show below that the net driving force along the pipe is

$$F_{\text{net}} = \rho g A \left(h_c(t) - h(L)\right), \tag{4}$$

where $h(L)$ is the height of the outlet at the current front position. Applying Newton's second law $F = m\,\ddot{L}$ gives

$$\rho A L\,\ddot{L} = \rho g A \left(h_c(t) - h(L)\right),$$

so, after cancellation,

$$L\,\ddot{L} = g\left(h_c(t) - h(L)\right),$$

and hence the basic lossless differential equation during the filling phase is

$$\ddot{L} = \frac{g}{L}\left(h_c(t) - h(L)\right), \qquad L(t_0) < L(t) < L_{\max}, \tag{5}$$

with $h(L)$ given by the piecewise definition above and $h_c(t)$ given by eq. (1). Physically, eq. (5) states that the acceleration $\ddot{L}$ of the water slug is driven by the instantaneous head difference $h_c(t) - h(L)$ and is inversely proportional to L, since a longer slug has more mass to accelerate.

The filling phase ends at time t_* such that $L(t_*) = L_{\max}$. Beyond this point, L stays fixed and we must use a different set of equations (see Part II).

1.5 Detailed Derivation of the Net Driving Force

We now derive F_{net} more explicitly by decomposing the contributions of pressure and gravity.

Local Force Balance on a Differential Element

Introduce a coordinate s along the pipe, measured from the inlet (we will later identify s with L at the slug front). Consider a small fluid element of length $\mathrm{d}s$ and cross-sectional area A. Its volume is $A\,\mathrm{d}s$ and its mass is

$$\mathrm{d}m = \rho A\,\mathrm{d}s.$$

Let $p(s,t)$ be the pressure and $h(s)$ the height of the centerline at position s. The tangent to the pipe makes a local angle $\theta(s)$ with the horizontal, so that

$$\sin\theta(s) = \frac{\mathrm{d}h}{\mathrm{d}s}.$$

The component of the weight along the pipe (positive in the direction of increasing s) is then

$$\mathrm{d}F_g = -\rho g A \sin\theta(s)\,\mathrm{d}s = -\rho g A\,\frac{\mathrm{d}h}{\mathrm{d}s}\,\mathrm{d}s.$$

For this element, the forces in the tangential direction (taken as positive in the direction of increasing s) are:

- Pressure on the upstream face (at s): $p(s,t)\,A$;
- Pressure on the downstream face (at $s+\mathrm{d}s$): $-p(s+\mathrm{d}s,t)\,A$;
- Component of weight along the pipe:

$$\mathrm{d}F_g = -\rho g A\,\frac{\mathrm{d}h}{\mathrm{d}s}\,\mathrm{d}s.$$

Thus the total tangential force on the element is

$$\mathrm{d}F = -\frac{\partial p}{\partial s}A\,\mathrm{d}s - \rho g A\frac{\mathrm{d}h}{\mathrm{d}s}\,\mathrm{d}s.$$

If the mean velocity in the slug is $v(t) = \dot{L}(t)$ and is uniform along the slug, then $\partial v/\partial s = 0$ and the local unsteady 1D momentum equation (Euler equation along a streamline) [1] reduces to

$$\rho\frac{\partial v}{\partial t} = -\frac{\partial p}{\partial s} - \rho g\frac{\mathrm{d}h}{\mathrm{d}s}.$$

Multiplying by $A\,\mathrm{d}s$ we recover

$$\mathrm{d}F = \rho A\,\mathrm{d}s\,\frac{\partial v}{\partial t} = \rho A\,\mathrm{d}s\,\frac{\mathrm{d}v}{\mathrm{d}t},$$

which is Newton's second law for the differential element.

Integration Along the Entire Water Slug

Integrate along the slug from the entrance at $s = 0$ (just inside the pipe) to the front of the slug at $s = L(t)$:

$$\begin{aligned} F_{\text{net}} &= \int_0^L \mathrm{d}F \\ &= \int_0^L \left(-\frac{\partial p}{\partial s}A - \rho g A\,\frac{\mathrm{d}h}{\mathrm{d}s}\right)\mathrm{d}s \\ &= -A\int_0^L \frac{\partial p}{\partial s}\,\mathrm{d}s - \rho g A\int_0^L \frac{\mathrm{d}h}{\mathrm{d}s}\,\mathrm{d}s. \end{aligned}$$

The integrals can be evaluated explicitly:

- Pressure contribution:

$$-A\int_0^L \frac{\partial p}{\partial s}\,\mathrm{d}s = -A\big(p(L,t)-p(0,t)\big) = A\big(p(0,t)-p(L,t)\big).$$

- Gravity (body force) contribution:

$$-\rho g A\int_0^L \frac{\mathrm{d}h}{\mathrm{d}s}\,\mathrm{d}s = -\rho g A\big(h(L)-h(0)\big) = \rho g A\big(h(0)-h(L)\big).$$

Therefore,

$$F_{\text{net}} = A\big(p(0,t)-p(L,t)\big) + \rho g A\big(h(0)-h(L)\big). \tag{6}$$

Relating End Pressures to the Container and Outlet

We now connect $p(0,t)$ and $p(L,t)$ to the free-surface and outlet conditions.

- At the container free surface ($z = h_c(t)$), the pressure is p_{atm}. Hydrostatic equilibrium in the container gives the pressure at the pipe inlet (which we call $p(0,t)$) as

$$p(0,t) = p_{\text{atm}} + \rho g\big(h_c(t)-h(0)\big),$$

  where $h(0)$ is the height of the pipe inlet.

- At the outlet, we assume the jet discharges to the atmosphere at height $h(L)$, so the pressure just at the outlet inside the pipe is

$$p(L,t) = p_{\text{atm}}.$$

Substituting into (6):

$$\begin{aligned} F_{\text{net}} &= A\left[p_{\text{atm}} + \rho g\big(h_c(t)-h(0)\big) - p_{\text{atm}}\right] + \rho g A\big(h(0)-h(L)\big) \\ &= \rho g A\big(h_c(t)-h(0)\big) + \rho g A\big(h(0)-h(L)\big) \\ &= \rho g A\big(h_c(t)-h(L)\big). \end{aligned}$$

Interpretation

From the above decomposition we can interpret

$$F_{\text{net}} = \underbrace{A\big(p(0,t)-p(L,t)\big)}_{\text{pressure drive}} + \underbrace{\rho g A\big(h(0)-h(L)\big)}_{\text{gravity drive}} = \rho g A\big(h_c(t)-h(L)\big).$$

In other words:

- The pressure difference between inlet and outlet, itself due to the water column in the container above the inlet, contributes a net push.
- The body force (weight) of the water along the pipe projects along the pipe centerline and adds another contribution.
- When these are combined, all dependence on the inlet height $h(0)$ cancels, and the net effect is equivalent to a single head difference between the free surface in the container at $h_c(t)$ and the outlet at $h(L)$.

Using $m = \rho A L$ and $F_{\text{net}} = m\,\ddot{L}$, we recover

$$\rho A L\,\ddot{L} = \rho g A\big(h_c(t)-h(L)\big),$$

which is precisely the lossless equation (5), now with the roles of pressure and gravity made explicit.

2 Part II: Loss Models and Phase-Split Dynamics

In this part we add distributed wall-friction losses (Darcy–Weisbach) and local (minor) losses, and we also distinguish explicitly between:

1. The *filling phase*, where the front advances and $v = \dot{L}$;
2. The *draining phase*, where the pipe is completely full and L is fixed, but $v(t)$ and $h_c(t)$ still evolve.

We denote the pipe diameter by D, and we continue to use $v(t)$ for the mean velocity in the pipe.

2.1 Kinematics and Phases

Filling Phase ($L(t_0) < L(t) < L_{\max}$)

During the filling phase, the front of the water plug advances along the pipe, so

$$v(t) = \dot{L}(t), \qquad L(t_0) < L(t) < L_{\max}. \tag{7}$$

The container height is given by eq. (1),

$$h_c(t) = h_c(t_0) - \frac{A}{A_0}\,[L(t) - L(t_0)].$$

The phase ends at time t_* when $L(t_*) = L_{\max}$.

Draining Phase ($t \geq t_*$, full pipe)

Once the water plug occupies the entire pipe, its length becomes fixed:

$$L(t) = L_{\max}, \qquad t \geq t_*,$$

and we must treat the mean velocity $v(t)$ as an independent state variable; in particular $v \neq \dot{L}$ since $\dot{L} = 0$.

The outlet height is now a constant

$$h_{\text{out}} = h(L_{\max}).$$

The volumetric flow rate is $Q(t) = Av(t)$, so the container volume and height evolve according to

$$\dot{h}_c(t) = -\frac{A}{A_0}\,v(t), \qquad t \geq t_*. \tag{8}$$

2.2 Distributed Friction: Darcy–Weisbach

The head loss due to wall friction over a length L of pipe is [1, 4]

$$h_f = f\,\frac{L}{D}\,\frac{v^2}{2g}, \tag{9}$$

where

- f is the Darcy friction factor (dimensionless),
- L is the wetted pipe length,
- D is the pipe diameter,

- v is the mean velocity in the pipe.

The Reynolds number is [1]

$$\mathrm{Re} = \frac{\rho v D}{\mu},$$

with ρ the density and μ the dynamic viscosity of the fluid.

For water at room temperature (approximately $\rho \approx 1000\,\mathrm{kg/m^3}$, $\mu \approx 1.0 \times 10^{-3}\,\mathrm{Pa \cdot s}$), typical regimes are:

- *Laminar flow* ($\mathrm{Re} \lesssim 2000$) [1]:

$$f = \frac{64}{\mathrm{Re}}.$$

- *Turbulent flow*: f depends on both Re and the relative roughness ε/D. A common correlation is Colebrook–White [2]:

$$\frac{1}{\sqrt{f}} = -2\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right),$$

  where ε is the roughness height.

Typical ranges of f for water in pipes:

- Smooth plastic tubing: $f \approx 0.015$–0.02 (high Re);
- Commercial steel pipe: $f \approx 0.02$–0.03.

Experimental Determination of f

In a straight pipe test section of known L and D, the pressure drop Δp is measured for a given flow rate (hence velocity v). The Darcy–Weisbach relation in pressure form is

$$\Delta p = f\,\frac{L}{D}\,\frac{\rho v^2}{2}.$$

Solving for f gives

$$f = \frac{2D\,\Delta p}{\rho v^2 L}.$$

Repeating this at various Re and roughnesses leads to the classical Moody diagram and empirical correlations [3].

2.3 Local (Minor) Losses

Each fitting or geometric disturbance is modeled as [1, 4]

$$h_m = K\,\frac{v^2}{2g}, \tag{10}$$

where K is a dimensionless loss coefficient determined by geometry.

If several such elements are present along the flow path, the total minor-loss head is

$$h_{\mathrm{minor,tot}} = \left(\sum_i K_i\right)\frac{v^2}{2g}.$$

Typical values of K (orders of magnitude) for water are [1, 4]:

- Sharp-edged entrance from a reservoir: $K \approx 0.5$;

- Well-rounded entrance: $K \approx 0.04$;
- Sudden exit to a large tank / atmosphere: $K \approx 1.0$;
- Smooth 90° bend: $K \approx 0.2$–0.5;
- Tight 90° elbow: $K \approx 0.5$–1.5;
- Fully open gate valve: $K \approx 0.1$–0.2;
- Fully open globe valve: $K \approx 8$–10.

Experimental Determination of K

A test loop is set up with the fitting of interest. The extra pressure drop Δp_{extra} across the fitting is measured, with straight-pipe losses subtracted out. Converting the extra drop to head,

$$h_m = \frac{\Delta p_{\text{extra}}}{\rho g},$$

the loss coefficient is

$$K = \frac{2g\, h_m}{v^2}, \qquad v = \frac{Q}{A},$$

where Q is the volumetric flow rate and A the pipe cross-sectional area.

2.4 Total Head Loss and Force

In the slug-flow model, over a wetted length L we take the total head loss as

$$h_{\text{loss}}(v, L) = f\,\frac{L}{D}\,\frac{v^2}{2g} + \left(\sum_i K_i\right)\frac{v^2}{2g}. \tag{11}$$

The corresponding resistive force (always opposing motion) is

$$F_{\text{loss}} = \rho g A\, h_{\text{loss}}(v, L)\ \operatorname{sgn}(v), \tag{12}$$

where $\operatorname{sgn}(v)$ is the sign of the velocity. Since $v\,\operatorname{sgn}(v) = |v|$, we can equivalently write the combination $|v|v$.

2.5 Extended Equation of Motion: Filling Phase

The momentum balance with pressure/gravity driving and losses for the *filling phase* is

$$\rho A L\,\ddot{L} = \rho g A\big(h_c(t) - h(L)\big) - \rho g A\, h_{\text{loss}}(\dot{L}, L)\ \operatorname{sgn}(\dot{L}).$$

Substituting

$$h_{\text{loss}}(\dot{L}, L) = \left(f\,\frac{L}{D} + \sum_i K_i\right)\frac{\dot{L}^2}{2g},$$

and cancelling ρA gives

$$L\,\ddot{L} = g\big(h_c(t) - h(L)\big) - \left(f\,\frac{L}{D} + \sum_i K_i\right)\frac{|\dot{L}|\dot{L}}{2}.$$

Thus the extended filling-phase ODE is

$$\ddot{L} = \frac{g}{L}\big(h_c(t) - h(L)\big) - \frac{1}{2L}\left(f\,\frac{L}{D} + \sum_i K_i\right)|\dot{L}|\dot{L}, \qquad L(t_0) < L(t) < L_{\max}. \tag{13}$$

Here:

- $h(L)$ is the geometric height along the pipe,
- $h_c(t)$ is given by eq. (1),
- f is the Darcy friction factor,
- K_i are the local loss coefficients.

When $f = 0$ and all $K_i = 0$, eq. (13) reduces to the lossless form (5). Physically, eq. (13) balances the driving head $h_c(t) - h(L)$ against a resistive term that grows like $|\dot{L}|\dot{L}$, i.e. quadratically with the mean velocity.

2.6 Extended Equation of Motion: Draining Phase

For the *draining phase* ($t \geq t_*$) the water occupies the entire pipe, so $L(t) = L_{\max}$ is constant and the velocity $v(t)$ is the dynamical variable. The outlet height is $h_{\text{out}} = h(L_{\max})$.

The momentum balance for the full slug of length $L_{\max}$ is

$$\rho A L_{\max}\, \dot{v} = \rho g A \big(h_c(t) - h_{\text{out}}\big) - \rho g A\, h_{\text{loss}}(v, L_{\max})\, \operatorname{sgn}(v),$$

with

$$h_{\text{loss}}(v, L_{\max}) = \left(f\, \frac{L_{\max}}{D} + \sum_i K_i \right) \frac{v^2}{2g}.$$

Cancelling ρA gives

$$L_{\max}\, \dot{v} = g\big(h_c(t) - h_{\text{out}}\big) - \left(f\, \frac{L_{\max}}{D} + \sum_i K_i \right) \frac{|v|v}{2}.$$

Together with eq. (8), the draining-phase dynamics are

$$\boxed{\begin{aligned} \dot{v} &= \frac{g}{L_{\max}}\big(h_c(t) - h_{\text{out}}\big) - \frac{1}{2L_{\max}} \left(f\, \frac{L_{\max}}{D} + \sum_i K_i \right) |v|v, \\ \dot{h}_c(t) &= -\,\frac{A}{A_0}\, v(t), \end{aligned}} \tag{14}$$

for $t \geq t_*$, where t_* is the time at which $L(t_*) = L_{\max}$. Physically, the first line of eq. (14) balances the driving head $h_c(t) - h_{\text{out}}$ against a quadratic friction and minor-loss term, while the second line expresses mass conservation for the container: the free-surface level drops at a rate proportional to the pipe flow rate.

The initial conditions at the start of the draining phase are

$$v(t_*) = \dot{L}(t_*), \qquad h_c(t_*) \text{ given by eq. (1) evaluated at } t_*.$$

In practice, one integrates the filling-phase ODE (13) from t_0 until $L(t) = L_{\max}$, then switches to the draining system (14) with these initial values. The flow eventually ceases as $h_c(t)$ approaches h_{out} and friction damps $v(t)$ to zero. For numerical work, these ODEs can be integrated with standard explicit time-stepping schemes (for example, a classical fourth-order Runge–Kutta method) [10].

Nondimensionalisation and Key Parameters

It is often useful to rewrite the draining-phase system in dimensionless form to highlight the combinations of parameters that control the dynamics. Let the reference length be the total pipe length $L_{\max}$ and define a gravitational time scale

$$T_0 = \sqrt{\frac{L_{\max}}{g}}, \qquad V_0 = \frac{L_{\max}}{T_0} = \sqrt{gL_{\max}}.$$

We also introduce a characteristic head difference, for example the initial driving head at the start of the draining phase,

$$H_0 = h_c(t_*) - h_{\text{out}},$$

and dimensionless variables

$$\hat{t} = \frac{t}{T_0}, \qquad \hat{v} = \frac{v}{V_0}, \qquad \hat{h}(\hat{t}) = \frac{h_c(t) - h_{\text{out}}}{H_0}.$$

In terms of these variables, eqs. (14) become

$$\frac{\mathrm{d}\hat{v}}{\mathrm{d}\hat{t}} = \frac{H_0}{L_{\max}}\hat{h} - \frac{1}{2}\left(f\,\frac{L_{\max}}{D} + \sum_i K_i\right)|\hat{v}|\hat{v}, \tag{15}$$

$$\frac{\mathrm{d}\hat{h}}{\mathrm{d}\hat{t}} = -\,\frac{A}{A_0}\,\frac{L_{\max}}{H_0}\,\hat{v}. \tag{16}$$

Thus the behaviour in the draining phase is governed mainly by the dimensionless groups

$$\frac{A}{A_0}, \qquad f\,\frac{L_{\max}}{D}, \qquad \sum_i K_i, \qquad \frac{H_0}{L_{\max}},$$

so experiments that share these combinations (even if the individual dimensional parameters differ) will exhibit dynamically similar siphon-draining behaviour.

Summary of Variables and Parameters

- State variables:
  - Filling phase: $L(t)$, $\dot{L}(t)$, $h_c(t)$.
  - Draining phase: $v(t)$, $h_c(t)$.
- Geometry:
  - Node coordinates: $(0, h_0)$, (x_1, h_1), …, (x_4, h_4).
  - Segment lengths: $L_1, \ldots, L_4$, total $L_{\max}$.
  - Height profile: $h(L)$ via segment maps h_i.
  - Outlet height: $h_{\text{out}} = h(L_{\max})$.
- Areas and diameter:
  - Pipe area: A; container free-surface area: A_0; pipe diameter: D.
- Loss parameters:
  - Darcy friction factor: f; minor-loss coefficients: K_i.
- Fluid properties and constants:
  - Density: ρ; dynamic viscosity: μ; gravity: g.

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References

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